



Johannes Carmesin

CURRICULUM VITAE

Personal details

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Current position

2024– Full Professor (W3-Chair in Discrete Structures)
Faculty of Mathematics and Computer Science, TU Freiberg, Germany, tenured

Previous positions

- 2022–2023 Head of Group in Combinatorics, Probability, and Algorithms
Leading a team of 8 permanent staff members, School of Mathematics, Univ. of Birmingham
- 2021–2024 Reader (Associate Professor)
School of Mathematics, University of Birmingham, tenured
- 2018–2021 Birmingham Fellowship (Tenure-Track)
Permanent research position, School of Mathematics, University of Birmingham
- 2015–2018 Junior Research Fellowship
Independent researcher, DPMMS, Emmanuel College, Cambridge, UK

Peer recognition

Selected research grants and awards

- Frontiers in Science Award (Beijing, 2026)
- European Prize in Combinatorics (Prague, 2023)
- EPSRC Early Career Fellowship (Birmingham, 2020–2024, PI, £1M, focused on the foundation of 3-dimensional graph minors theory)

Selected Plenary Lectures

1. 47th Australasian Combinatorics Conference (2025), Wellington, New Zealand
2. CANADAM (2019), 400 participants, Vancouver, Canada
3. Oberwolfach Workshop on Graph Theory (2019), Opening Lecture, Oberwolfach, Germany
4. Structure in Graphs and Matroids (2017), Tutte Centennial Conference, Waterloo, Canada

All the above are plenary lectures of approximately 50 minutes in duration. In total I have delivered well above 10 plenary lectures at international conferences and workshops, beyond the above these include: Eurocomb, Kolkom, Matroid Union Seminar, London Colloquia in Combinatorics, Dimeia in Czech Republic, Higher Infinity Workshop in Cambridge.

Editorships

- Associate Editor for the journal *Innovations in Graph Theory* (since 2023; open access and non-commercial).
- Associate Editor for the journal *Discrete Mathematics* (since 2019).



Research achievements

My research in Combinatorics builds structural methods that interact deeply with neighbouring areas of mathematics. Some of my contributions engage with or build on ideas from topology, differential geometry of 3-manifolds, algebra (matroid theory), and algorithms. This cross-disciplinary perspective has enabled progress on long-standing problems and opened new directions, including combinatorial viewpoints on phenomena related to Perelman's work. Fourteen of my papers have appeared in *Combinatorica* or *JCTB*, and I have written about 50 papers in total. My work has also appeared in *Memoirs of the American Mathematical Society* and *FOCS*.

1. EMBEDDING SIMPLY CONNECTED 2-COMPLEXES IN 3-SPACE, *Memoirs of the American Mathematical Society*, 319 (2026), no. 1625.

This work characterizes the embeddability of simply connected 2-dimensional simplicial complexes in 3-space, analogous to Kuratowski's characterization of graph planarity. It addresses key questions posed by Abel Prize winner Lovász, geometer Pardon, and combinatorialist Wagner. Using methods from Topology, Differential Geometry of 3-manifolds, Algebra (Matroid Theory), and Combinatorics, we present a quadratic time algorithm for embedding verification. These results were presented in the Opening Lecture at the Oberwolfach Workshop and inspired further work in my group and beyond.

2. CANONICAL DECOMPOSITIONS OF 3-CONNECTED GRAPHS (with J. Kurkofka), *IEEE Symposium on Foundations of Computer Science (FOCS)*, 2023, pages 1887–1920; journal version: *Advances in Combinatorics*, 2025:7, 73 pages.

One of the most fundamental tools in graph theory is the theory of connectivity, which decomposes a graph into smaller pieces, solving the problem of interest locally before combining these solutions globally. While the theory of separators of size one or two has been well understood since the 1960s due to Tutte's Theorem, it was long believed that an extension to separators of size three was impossible. In this paper, we introduce a new perspective that allows such an extension, impacting both Group Theory and Computer Science.

3. LOCAL 2-SEPARATORS, *Journal of Combinatorial Theory, Series B*, 2022, 156:101–144.

The emergence of Data Science has led to new challenges for Graph Theory, where algorithms need to be faster and scale efficiently in large networks. Traditionally, graph algorithms focused on polynomial time solutions, but recent applications require algorithms that run independently of the number of vertices, often using parallel computing. In this paper, I refine the theory of graph connectivity to study locally separating vertex sets, leading to applications in Group Theory, Graph Theory, and Large Networks.

4. A GRAPH MINORS APPROACH TO TEMPORAL SEQUENCES (with W. Turner), *ACM Symposium on Theory of Computing (STOC)*, 2026, pages 1388–1396. Full version: [arXiv:2504.00704](https://arxiv.org/abs/2504.00704), 70 pages.

We develop a structural approach to simultaneous embeddability in temporal sequences of graphs, inspired by graph minor theory. Our main result is a classification theorem for 2-connected temporal sequences: we identify five obstruction classes and show that every 2-connected temporal sequence is either simultaneously embeddable or admits a sequence of improvements leading to an obstruction. This structural insight leads to a polynomial-time algorithm for deciding the simultaneous embeddability of 2-connected temporal sequences. The restriction to 2-connected sequences is necessary, as the problem is NP-hard for connected graphs, while trivial for 3-connected graphs. As a consequence, our framework also resolves the rooted-tree SEFE problem, a natural extension of the well-studied sunflower SEFE. More broadly, our results demonstrate the applicability of graph minor techniques to evolving graph structures and provide a foundation for future algorithmic and structural investigations in temporal graph theory.



5. Dual matroids of 2-complexes—Revisited, *Bulletin of the London Mathematical Society*, 2025, 57(7):2033–2044.
We extend Whitney’s characterisation of planar duality to embeddings of 2-complexes in 3-space. The paper introduces a new geometric representation of matroids via dual 2-complexes. Matroids representable by 2-complexes form a natural extension of the class of graphic matroids.
6. Characterising graphs with no subdivision of a wheel of bounded diameter, *Journal of Combinatorial Theory, Series B*, 2023, 161:21–51.
This paper is a key application of our local-separators framework (see Paper 3). The sparsity theory of Nešetřil and Ossona de Mendez is remarkably general and unifies many equivalent notions of sparse graph classes. We demonstrate, perhaps surprisingly, that this so far fairly abstract theory allows for explicit structural computations reminiscent of the characterisation theorems in graph minor theory. Concretely, we show that excluding an r -bounded wheel—a particular type of shallow minor with branch sets of diameter at most r —is equivalent to admitting graph decompositions of locality r and width 2.
7. On Andreae’s Ubiquity Conjecture, *Journal of Combinatorial Theory, Series B*, 2023, 162:68–70.
We give a construction showing that Andreae’s Ubiquity Conjecture needs to be amended by adding a natural extra assumption.
8. All graphs have tree-decompositions displaying their topological ends, *Combinatorica*, 2019, 39:545–596.
A main theme in graph theory is to approximate infinite graphs by trees, which then unlocks a wide range of proof techniques. This paper proves Halin’s 1964 End-faithful Spanning Tree Conjecture in an amended form: every infinite graph admits a single spanning tree whose ends correspond bijectively to its topological ends. This shows that the global ‘shape’ of an infinite graph can be captured by one canonical tree. The result complements a construction of Thomas and Seymour, who proved that topological ends cannot in general be replaced by vertex-ends.
9. Connectivity and tree-structure in finite graphs (with R. Diestel, F. Hundertmark & M. Stein), *Combinatorica*, 2014, 34:11–46.
This paper proves a substantial generalisation of the tangle–tree theorem of Robertson and Seymour.
10. GRAPH THEORY—A SURVEY ON THE OCCASION OF THE ABEL PRIZE FOR LÁSZLÓ LOVÁSZ, *Jahresbericht der Deutschen Mathematiker-Vereinigung*, 2022, 124:83–108.
This invited article surveys major challenges in graph theory for the coming decades, prepared in honour of the Abel Prize awarded to L. Lovász.



Education

- 2019–2021 Postgraduate Certificate in Higher Education (PGCHE)
With distinction, School of Mathematics, University of Birmingham, UK
- 2018 Habilitation in Mathematics
Department of Mathematics, University of Hamburg, Germany
- 30/11/2015 PhD: Scholar of the Studienstiftung des deutschen Volkes
Awarded *summa cum laude*, Department of Mathematics, University of Hamburg, Germany
Supervisor: Reinhard Diestel
- 2012 Master's degree (studies 2007–2012)
Completed half a year early, with distinction, including five published papers,
Department of Mathematics, University of Hamburg, Germany
- 2005 – 2009 Compensatory military service at 'Technisches Hilfswerk'

Other contributions to the research community

Selected Conference Organisation

- Bertinoro Workshop on Algorithms and Graphs (2025) Co-organized with Wollan and Bonamy, focusing on the intersection of algorithms and graph theory. I initiated and funded the re-launch of this event after it had not taken place for a few years.
- Ukrainian Summer School in Combinatorics in Freiberg (2025) Co-organized with Pikhurko and Kroeker, offering advanced training for PhD students and early-career researchers. The Ukrainian mathematicians Nekrashevych and Radchenko as well as Bonamy and Král' delivered mini-courses, and we awarded the Ukrainian Prize in Combinatorics to Andrii Arman.

Public Engagement

- Initiator of *Probe Uni* (since 2025 annual event): allows school students to come to university for a week to attend lectures, particularly well-received for pupils without academic background.
- Public lecture as part of the series *Mathematik im Gespräch* at the Erlebnisland Mathematik, Dresden (2025).
- Workshops for high school students as part of the "Talentförderung Mathematik" programme (2006–2012). Delivered 4-hour workshops every two weeks and coordinated the programme in the Harburg area.

Teaching

- Coordinator of the teaching activities in Discrete Mathematics and Theoretical Computer Science at TU Freiberg.
- Designed and delivered courses such as Discrete Structures I+II, Graph Theory I+II, Complexity Theory, Algorithmic Topology.

Selected supervision, reviewing and administration

- Two PhD completions. I currently supervise two postdocs and four PhD students.
- Chair of two EPSRC Panels (2023); panel member in 2021.
- Outstanding reviewer (EPSRC, JGT); reviews: ≈ 40 grants, ≈ 100 papers, ≈ 10 PhD theses.
- Coorganiser of the weekly Freiberg Research Seminar in Discrete Mathematics and Algebra.